

Behavioral Authentication with Head-Tilt Based Locomotion for Metaverse

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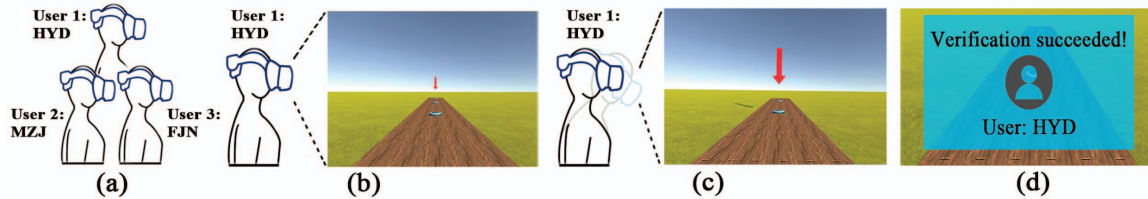


Figure 1: Illustration of behavioral authentication with head-tilt based locomotion. (a) Support multiple users for authentication. (b) Initial user viewpoint in VR without tilting the head. (c) Tilt the head to control the speed of the virtual viewpoint. (d) Behavioral authentication is performed when reaching to a target, and a user's name is displayed after successful authentication.

ABSTRACT

In order to address issues with user authentication such as shoulder surfing attacks and authentication timeouts, we proposed an implicit authentication method for locomotion in virtual reality (VR). The method used head-tilt based locomotion for user authentication. We conducted a study to collect a dataset of biological behavior for head-tilt based locomotion. Random Forest and Decision Tree algorithms were used to train the model, comparing the fitting effects of the two algorithms at different distances (3 meters, 6 meters, 9 meters, and 12 meters). We examined the model's accuracy and discovered that it had a high accuracy in identifying one of ten users (above 95 %).

Index Terms: Authentication—Behavioural Biometrics—Body-centric locomotion—Modeling

1 INTRODUCTION

With the rise of the metaverse, VR devices have been employed in production and everyday life. VR headsets are often connected to a wide range of private internet accounts, including social networks, email, and payment, which record our daily activities, spending, and health data [6]. The question of how to protect our privacy and security in VR has become a problem worthy of attention.

Using biometric behavior for user identification is an implicit authentication method. Everyone participates in an activity with their own distinct behaviors. In contrast to explicit user authentication, implicit user authentication can avoid the use of digital inputs like keys or passwords and prevent shoulder surfing attacks [5].

Body-centric locomotion for behavioral authentication is a continuous method. When users perform a task in VR, such as navigation, they frequently have to go from one location to another. Locomotion occurs continuously throughout the whole navigation process. Within such a context, behavioral biometric authentication can deal with issues like unexpected logouts and authentication timeouts [4].

Gait features for real walking have been used for authentication in previous works [6]. However, no work has used body-centric locomotion for authentication. Compared to real walking, body-centric locomotion has the advantage of moving in place in a small

space. The head-based locomotion method is a hands-free method for body-centric locomotion. Users can synchronize other tasks with their hands when moving in this way. Head motion is also considered to be a useful behavioral biometric in user authentication.

To solve the above issues, taking into account the benefits of body-centric locomotion and the accuracy of behavioral biometrics with different body parts, this paper proposes behavioral authentication with head-tilt based locomotion (Fig. 1). We will train a model to verify the effectiveness of the head-based locomotion method for VR user authentication at different moving distances.

2 RELATED WORK

2.1 Head-tilt based locomotion

Users can control the speed or direction of the virtual viewpoint by tilting or yawing their heads. Tregillus et al. [8] proposed a head-based locomotion method using head tilt. The results showed that there was no motion sickness when using head tilt for navigation tasks. Terziman et al. [7] proposed the shake-your-head method by tilting or translating the head to control movements such as turning, jumping, and crawling. This method outperformed keyboard and joystick in terms of fun, presence, and appreciation. Gao et al. [1] evaluated how different transfer functions and body parts affected the effectiveness and accuracy of locomotion in virtual reality. Compared to other body-centric locomotion methods, using the head to control both the direction and the speed of the virtual viewpoint resulted in the lowest rate of failed trials.

2.2 Behavioural Biometrics

Pfeuffer et al. [5] investigated which body movement behaviors were suitable for user authentication of pointing, grabbing, walking, and typing in VR. According to their results, the relative distance feature set between the dominant hand, non-dominant hand, and head had the best accuracy for identifying users when people were walking. The second most accurate feature was head movement. Additionally, Shen et al. [6] showed that head motion was appropriate for walking tasks. They discovered that head movement rather than hand movement was a better way to characterize a user's walking.

3 METHODOLOGY

3.1 Experiment

We designed a within-subject study to collect data on head-based locomotion in a virtual environment.

We recruited 10 participants (5 male and 5 female) from the local campus, aged between 19 and 25 years old. The average height of this group was 167.6 cm, with a height range of 155 cm to 178 cm.

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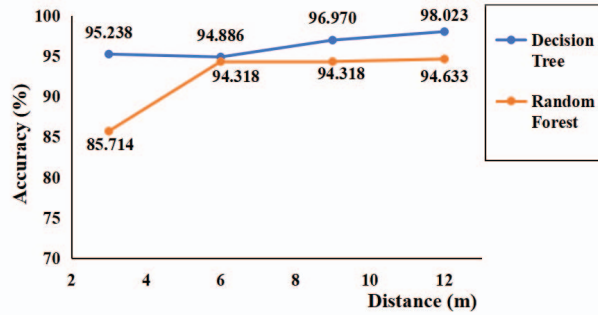


Figure 2: Accuracy comparison of random forest and decision tree methods with different movement distances.

Family members often use VR headsets together, and the maximum number of authorized users is usually five [6]. Therefore, using the data of 10 subjects to build the model is reasonable.

The experiment required collecting data on the head-mounted display's position, rotation angle, speed, and angular velocity. Each participant was required to take part in two data collection sessions, with the second session needing to be held more than three days after the first session [5], [6].

In the experiment scene, there were four targets on a straight path. The targets were 3 meters, 6 meters, 9 meters, and 12 meters from the starting position. A target was represented by a blue circle with a radius of 0.5 m, and the red arrow above the target indicated the target's location [1] (Fig. 1 (b) & (c)).

Participants were given a brief introduction, and asked to complete a questionnaire, and sign a consent form before the experiment began. During the practice phase, participants were informed about the process of the exercise task and were equipped with a VR headset and a controller. Participants were asked to move freely in the virtual scene for 5 minutes using the head-based method. The main goal of the practice was for participants to become familiar with this locomotion method in order to complete the tasks. During the formal testing phase, participants had to use the same locomotion method as in the practice phase. In each session, a participant needed to complete four trials. In each trial, one of the four targets was randomly selected as the destination for locomotion. Participants began a trial by pushing the controller's trigger button. They then went to find a target indicated by a red arrow. Participants pressed the controller's trigger for confirmation upon reaching the destination.

3.2 Modelling

The data from the first session was used to train the model, and the data from the second session was used to evaluate the model's accuracy. When users tilted their heads for locomotion, data was collected once for every 0.1-meter increase in the distance between the current position and the starting position. For training the model, we collected a total of 2925 pieces of data. We chose features related to head motion for training in order to better apply the head-tilt based method for locomotion to VR identity authentication. The features included head angular velocity, head rotation angle, height of the headset on the Y axis, distance of head motion, and speed of the virtual viewpoint [1], [5], [6], [2].

We trained the data with some classifiers using the Scikit-learn Machine Learning Library for Python, such as Support Vector Machine (SVM), K-Nearest Neighbor (KNN), Random Forest (RF), and Decision Tree (DT) [5], [3]. The accuracy of the SVM and KNN algorithms was poor due to the limited sample size of our data (10 users). However, the results of the random forests and decision trees are excellent. Then, we compared the results of Decision Tree and Random Forest at different distances for locomotion (3 meters, 6

meters, 9 meters, and 12 meters).

The results showed that both the decision tree method and the random forest method had excellent fitting effects, with the decision tree method having slightly higher accuracy at different distances (Fig. 2). The random forest method achieved 85.714 %, 94.318 %, 94.318 %, and 94.633 % accuracy at 3 meters, 6 meters, 9 meters, and 12 meters, respectively. The model's accuracy at 3 meters, 6 meters, 9 meters, and 12 meters, respectively, was 95.238 %, 94.886 %, 96.970 %, and 98.023 % using the decision tree method. The accuracy of the algorithm might be improved with the increase of distance.

3.3 Evaluation

We required participants to move a distance in the virtual environment for user authentication. When a user successfully completed the verification, the user interface displayed the name of the current user (Fig. 1 (d)). We used the evaluation method suggested by Pfeuffer et al. [5] to verify the accuracy of model classification. This metric uses the number of correct classification decisions divided by the total number of decisions. Our model could identify one out of every ten users with above 95 % accuracy.

4 CONCLUSION

In this work, we presented behavioral authentication with head-tilt based locomotion. Random forest and decision tree algorithms at different moving distances (3 meters, 6 meters, 9 meters, and 12 meters) were adopted and had a good fitting effect, and the accuracy of recognizing one of the ten users is above 95 %. This method may be applied to the metaverse with virtual navigation.

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